



RAMCO INSTITUTE OF TECHNOLOGY

Approved by AICTE, New Delhi and Affiliated to Anna University, Chennai

An ISO 9001: 2015 Certified Institution and Accreditation by NAAC

NBA Accredited Programs: CSE, ECE, EEE and Mechanical Engineering

DEPARTMENT OF MECHANICAL ENGINEERING

Presents

TECHNO MECHANIC

Annual Technical Magazine for
Mechanical Engineers

ISSUE 3

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RAMCO INSTITUTE OF TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

VISION, MISSION STATEMENTS

COLLEGE VISION

To evolve as an Institute of international repute in offering high-quality technical education, Research and extension programmes in order to create knowledgeable, professionally competent and skilled Engineers and Technologists capable of working in multi-disciplinary environment to cater to the societal needs.

COLLEGE MISSION

To accomplish its unique vision, the Institute has a far-reaching mission that aims:

- To offer higher education in Engineering and Technology with highest level of quality, Professionalism and ethical standards
- To equip the students with up-to date knowledge in cutting-edge technologies, wisdom, creativity and passion for innovation, and life-long learning skills
- To constantly motivate and involve the students and faculty members in the education process for continuously improving their performance to achieve excellence.

QUALITY POLICY

RIT aims to impart Quality Education in Engineering and Technology through an effective teaching-learning process, up-gradation of facilities and human resources, collaborating with Industry for promoting training and placement, research and consultancy activities with a commitment to continual improvement of Quality Management System.

DEPARTMENT VISION

Recognized nationally for its quality education, research and extension programmes in Mechanical Engineering.

DEPARTMENT MISSION

- Producing technically competent and socially responsible mechanical engineers by imparting knowledge of cutting edge technologies and contemporary issues.
- Encouraging students for active participation in Co-curricular and Extracurricular activities.
- Motivate students to pursue higher studies and research.
- Establish centres of excellence in promising fields of Mechanical Engineering.
- Develop linkages with industries, R&D organizations and leading educational institutions in India and abroad for achieving excellence in teaching, research and consultancy activities.

PROGRAM EDUCATIONAL OBJECTIVES (PEOS)

- Graduates will be able to apply the concept of mathematics, science and engineering to solve problems in mechanical engineering/related fields and acquire professional competence.
- Graduates will demonstrate professional development by pursuing higher education and professional certification in the areas of mechanical engineering.
- Graduates will function both independently and as a member of a team in accomplishment of project with social and ethical responsibility.

PROGRAM SPECIFIC OUTCOMES (PSOS)

- Design components of products and develop systems using CAD/CAE tools.
- Provide manufacturing solutions, including material, process(es), process plans and inspect products and their components.
- Design and analyse heat power (thermal) devices/system.
- Develop simulation models of thermal and manufacturing systems using Computer aided analysis tools.

EDITORIAL MESSAGE



HEAD OF THE DEPARTMENT

It is my pleasure to welcome you to the annual Mechanical Engineering Magazine of Ramco Institute of Technology. As the world continues to evolve and advance at an unprecedented pace, the field of mechanical engineering remains at the forefront of innovation and progress.

This year's magazine is filled with insightful articles, thought-provoking research, and inspiring stories from the students, faculty, and staff of our esteemed institution. From cutting-edge research in robotics and automation to the latest advancements in materials science and manufacturing, our mechanical engineering program is dedicated to exploring and pushing the boundaries of what is possible.

I would like to express my gratitude to all of our contributors for their hard work and dedication in making this magazine possible. I am confident that the insights and perspectives shared in these pages will inspire and inform readers from all backgrounds and disciplines.

As we look towards the future, the field of mechanical engineering will undoubtedly continue to play a critical role in shaping the world around us. At Ramco Institute of Technology, we are proud to be training the next generation of mechanical engineers who will lead the charge in tackling the world's most pressing challenges. Thank you for your continued support of our program, and I hope you enjoy this year's magazine.



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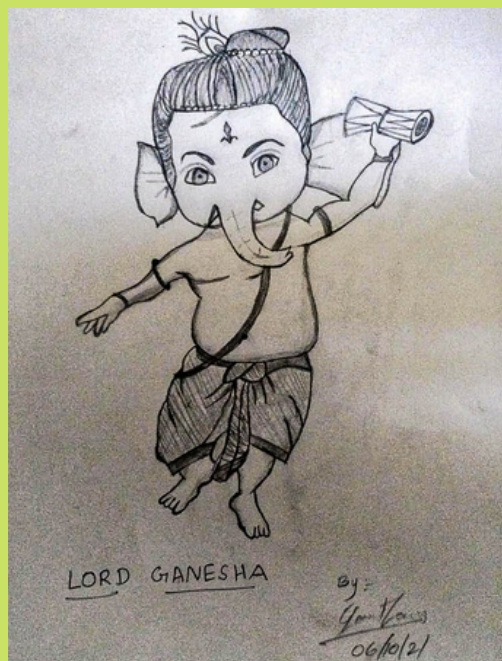
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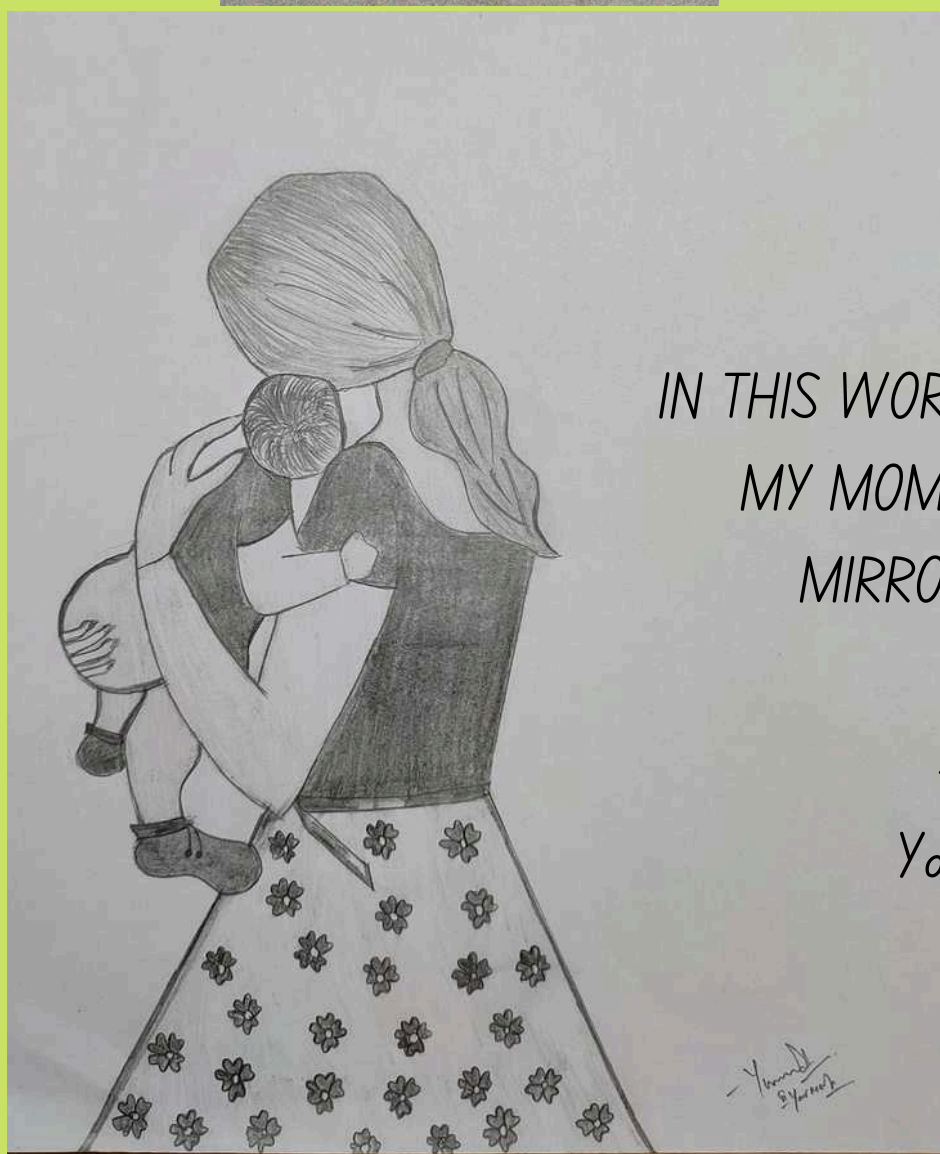
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Poem



J. GOWTHAM
BATCH 2021-2025



IN THIS WORLD OF CRITICISM,
MY MOM IS THE ONLY
MIRROR I TRUST...

粥

-Youn

Your Mer

-Younis Syam



YUVARAJ SK
BATCH 2020-2024

 [starky_vj](https://www.instagram.com/starky_vj)

கொரோனா போராளிகள்....

கண்ணீர் உலகம் அறிந்த மொழி,
உலகம் ஒன்றுகூடி அம்மொழியை பேசிக்கொண்டிருந்த சமயம்,
நம்பிக்கை காற்றில் கரைந்த தருணம்,
கேள்விகள் பல தோன்றின,
அக்கேள்விகளுக்கான ஒரே விடை
“ மருத்துவர்கள் ” என்பதே.....

நாட்கள் நிற்காமல் ஓடின குடும்பங்களை பிரிந்த நாள் முதல் அவர்களுக்கு,
அரவணைப்பை தந்த அம்மாவின் முகம் தெரிந்தது அக்குழந்தைக்கு சில அடி தொலைவில்,
பேசும் சத்தம் கேட்காத தொலைவில் தாயும் சேயும் பேசினர் இருவரின் கண்ணீரால்,
அக்கண்ணீர் கூறியது அத்தாய் மருத்துவரின் அர்ப்பணிப்பை.

காவலுக்கு சென்ற அப்பாவை காண துடிக்கும் மகனின் கண்கள் கூறின,
நிற்க தயங்காத கால்கள் கூறின,
தூங்க உதவிய சாலைகளும் மேசைகளும்,
அதற்கு சாட்சியாய் நின்ற மின்கம்பங்களும் கூறின,
பிறர் என்று எண்ணி பார்க்காத அவர்களின் மனம் கூறியது “காவலர்களின்” அர்ப்பணிப்பை.

அசுத்தம் அடையும் கைகளை மறந்த எண்ணம் கூறியது,
முகத்தில் உள்ள முக கவசத்தின் அச்சுகள் கூறியது,
தன் நாடு என் வீடு என்ற மனம் படைத்த அவர்களின் குணம் கூறியது,
“துப்புறவு தூய்மையாளர்களின்” அர்ப்பணிப்பை.

அம்மா.....

கருவில் தோன்றிய நாள் முதல் வலிகள் பல தோன்றினாலும் மறைத்து சிரிக்கிறாய்,
நித்தம் எனக்காக.....
நீ மறைப்பதை அறிந்த என் மனம் சிரிப்பதே உனக்காக.....
பாசத்திற்கு பஞ்சமில்லை,
அக்கரைக்கு அளவில்லை,
ஆனாலும் கண்டிப்பு குறைந்ததில்லை.....
கண்டிக்கும் உன் மனமோ அறிந்ததில்லை என்னை காயப்படுத்த.....
உலகம் சொல்லிக்கொடுத்தாய் என் கரங்களைப் பிடித்து,
நான் அறிந்ததோ என் உலகம் உன்னைப்பற்றி.....

காதல்.....

பகல் இரவாய் என் நினைவில் வாழும் நிஜமே,
உன் நிழலாகிய என்னுடன்,
நிலவினை தேடிச்செல்லும் பயணத்தில்,
உயர்ந்து நிற்கும் மலைகள் கடந்து,
கடலின் அலை செல்லும் திசையில் செல்வோம்.
நிலவினை அடையும் அந்நாள்,
உன் நிழலாகிய நான் ...உன் கை கோர்த்து நிஜமென மறுவேன்,
உன்னவனாக

-சந்தான கிருஷ்ணன்.ஜெ
Batch 2020-2024

PHOTOGRAPHY



***SRI SABARI VENKATESH
BATCH 2021-2025***

CRISIS OF OUR TIMES

Mr. A. Parthiban A, Batch 2020-2024



What do you think is the crisis of our time? The answer is climate change. The climate change started when there was **INDUSTRIAL REVOLUTION**. Aka **IR (1.0)** it happened in (1760-1830). We peoples started to develop as a society and also started to emit **GHG** Green House Gases such as carbon-di-oxide (CO₂), methane (CH₄) etc... as much as possible. Then happened the IR (2.0) from (1830-1920) which made it worst. Then the government of the SO CALLED developed countries realized it, but when? In 1990 when India was in deep financial crisis. In the form of Kyoto protocol. All the developed nations made all the damage to the environment and asking the developing and under developed countries to take repair it, By the IR (3.0) (1950-2010) made everything impossible when it comes to environment. Any how the Kyoto protocol was not implemented successfully even by the EU. When the world is ready for next IR in 2015 then POTUS Barackobama along with many prominent countries signed the agreement known as (**Paris pact**). But the successor of Mr.Barack aka (DJT) Donald John Trump, withdraws from Paris agreement which made whole world laugh at USA. Then POTUS DJT don't even regret for his activity. This Paris agreement led to the COP Conference of parties which India attended recently. What is climate change? The unnatural harms which caused by natural disasters, meaning the nature is giving back the pain to the humans. Till 2099 the world can be heated to 1.5 degree Celsius. To attain this target we have a plan named carbon budgeting which basically means the sharing of emission within the great emitters of GHG like USA, CHINA, RUSSIA, EU, BRAZIL and INDIA. But when it comes to India we have the 2nd largest population in the world but we are emitting just 7.5% of world's emission so here is an uneven distribution. As a developing nation India is a largest producer of renewable source of energy, the developed countries should learn from India. I am not saying that India shouldn't contribute for the climate change it shall have some allowances in this mission since India didn't signed any agreement regarding the climate change in Glasgow submit but India have pledged that its mission to achieve NET 0 emission by 2070, even it's a numerical target India's progress is way better than the western countries. Can India achieve Net 0 in 2070? Let's have a look at India's population 1.37 billion, and the active vehicle count as per CNN repot 297 million, 79.5% of total electricity produced in the country are from non-renewable energy sources till there are tribal population in India who doesn't even know what is electricity, with these many obstacles can India do that.

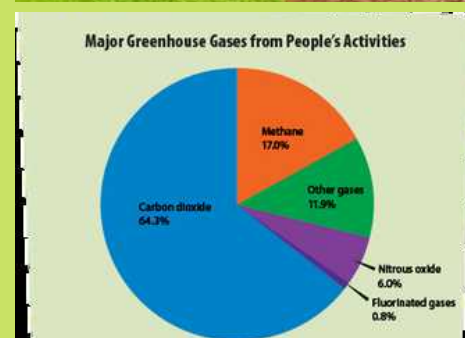
ROAD MAP

- 500 gigawatts of non-fossil fuel.
- 50% of renewable source of energy.
- 1 billion tons carbon emission.
- 45% carbon emissions.
- Carbon net zero by 2070.
- This is the plan which India has right now which looks completely good, as an emerging economy can India act on this plan immediately. For this the country needs humungous funding, many relay on this coal processing and power production. For that global financing approved \$15.3 billion to the emerging economy nearly \$1 billion has got approval for India's funding. The climate scientist and geographers says that India should not believe on the marathon it should achieve a major part of the mission on 2030. They say we have no control on it but we haven't loosened on 2030. As like India all other nations which are greater polluters should also have to work sincerely on their targets. If that happens no force on earth can stop achieving it.

What can be the contribution from common man?

- Planting and maintaining a tree till it becomes Aathmanirbar.
- Reducing the usage of SUP Single Use Plastics.
- Avoiding the usage of vehicles for shorter distance.

This is how we people can contribute to curb the crisis. Government of India is confident in achieving in it let's help the mission how we have done in the past while facing the pandemic. The main objective is to create awareness of climate change among the students. The future citizens of the world's largest democracy should be proud that our before generation is a visionary.



INDUSTRIAL REVOLUTION

QUOTES & POEM



P.R. JAYA SURYA
BATCH 2020-2024

 [surya_official_93](https://www.instagram.com/surya_official_93)

*If you are tall,
People ask, why are you tall?*

*If you are dark,
People ask, why are you dark?*

*If you are fair,
People ask, why are you fair?*

*If you are Skinny,
People ask, why are you Skinny?*

*If you are shout,
People ask, why are you shout?*

*At the End of the day you will be judged no matter what Live your life the want
to... way*

*When Life Gives You A Hundred Reason To Cry, Show Life That You Have
Thousand Reasons
To Smile....*

M. SUBBIAH@SELVAM
BATCH 2021-2025

 [subbiahselvam8](https://www.instagram.com/subbiahselvam8)



AN INTENSIVE ANALYSIS OF BHOPAL GAS TRAGEDY -WORLD'S WORST INDUSTRIAL DISASTER

Mr. M. Thojesh Nandha, Batch (2018-2022)

 [Thojesh Nandha](#)

 [thojesh_nandha](#)



Industrial safety refers to the regulations and safeguards put in place to protect plant and factory workers from dangers that might result in injury. Industrial safety is required to examine all potential causes of accidents in order to prevent death or permanent disability of any industrial employee, as well as damage to machinery and materials. It is required to eliminate work stoppages and production losses caused by accidents.



Union Carbide India Limited was founded in the year 1934. Batteries carbon products, welding equipment, plastics, industrial chemicals, pesticides, marine products. Union Carbide and Carbon Corporation (UCC), U.S-based, owns 50.9 percent of UCIL, while Indian investors, notably the Government of India and government-controlled banks, owned 49.1 percent. The UCIL factory was built in 1969 to produce the pesticide Sevin (UCC's brand name for carbaryl) using methyl isocyanate (MIC) as an intermediate. An MIC production plant was added to the UCIL site in 1979.

- Three underground 68,000-litre (18,000 gal) liquid MIC storage tanks were situated in the Bhopal UCIL facility: E610, E611, and E619. According to UCC safety requirements, no one tank should be loaded more than 50% (in this case, 30 tonnes) with liquid MIC.
- Inert nitrogen gas was used to pressurize all the tanks. Water was suspected to have entered a side pipe and into Tank E610 during the late evening hours of December 2, 1984, while trying to clear it, which housed the 42 tonnes of MIC that had been there since late October.
- The reaction in tank E610 had reached a critical condition at an alarming rate by 12:40 a.m., and the pressure in the tank had reached 40 psi (275.8 kPa).
- Despite the fact that atmospheric venting of deadly MIC gas had already commenced, a concrete slab above tank E610 cracked when the emergency relief valve broke open, and pressure in the tank continued to rise to 55 psi (379.2 kPa).
- In 45 to 60 minutes, about 30 tonnes of MIC were released into the atmosphere from the tank. Within two hours, this would climb to 40 tonnes. Over Bhopal, the fumes were blown in a south-easterly direction.



Conclusion

Approximately 45 tonnes of the hazardous chemical methyl isocyanate escaped from an insecticide facility controlled by Union Carbide Corporation's Indian division. The gas spread quickly across the heavily populated areas surrounding the factory, killing hundreds of people and causing panic as tens of thousands more rushed to evacuate Bhopal. The death was believed to reach between 15,000 and 20,000 people in the end. A half-million survivors had lung issues, eye irritation or blindness, and other illnesses as a result of their exposure to the deadly gas, with many receiving only a few hundred dollars in compensation.

AN INDUSTRIAL CASE STUDY ON TOTAL PRODUCTIVE MAINTENANCE (TPM)

Total Productive Maintenance:

TPM (Total Productive Maintenance) is a holistic approach to equipment maintenance that strives to achieve perfect production. Total Productive Maintenance (TPM) can be considered as the medical science of machines. A company-wide team-based effort to build quality into equipment and to improve overall equipment effectiveness. The main objective of the TPM is to markedly increase production while at same time, increasing employee morale and job satisfaction.

Goals of Lean Manufacturing:

- No Breakdowns
- No Small Stops or Slow Running
- No Defects
- In addition, it values a safe working environment:
- No Accidents



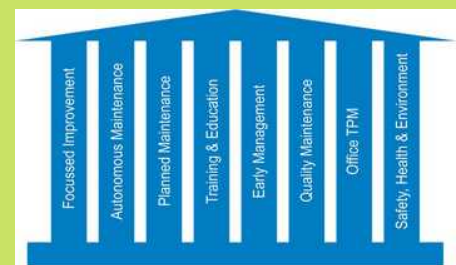
Case Study:

In 1931, Hindustan Vanaspati Manufacturing became the company's first Indian subsidiary. In 1933, Lever Brothers India Limited was founded. In 1935, United Traders Limited was founded. In 1956, the three firms united to become HILL. Hindustan Unilever Limited is primarily owned by the Anglo-Dutch business Unilever. In June 2007, the company was renamed Hindustan Unilever Ltd.

Food Brands, Home care Brands, Personal care Brands and Water

The three best-known process improvement methods are **(1) TPM is used to strengthen the stability of machinery, (2) Lean is used to boost the value stream towards the client, and (3) Six Sigma is used to eliminate process variation.**

Total Productive Maintenance (TPM), a 20-year-old improvement approach pioneered by Unilever, focuses on enhancing productivity. Machine-stoppages and other losses, such as sub-standard quality, are recognised and corrected using the Overall Equipment Effectiveness (OEE) measurement. TPM is generally used when the machinery is expensive and it is critical to get it up and running quickly. TPM gained a lot of traction, especially in the food business.



20 Years TPM Within Unilever:

- Unilever began implementing Total Productive Maintenance (TPM) in 1996, over 20 years ago. (TPM).
- Since 2008, Unilever also produces peanut butter in their factory in Rotterdam.
- At Silvassa factories of HUL, Worn out machine parts were stored and displayed at Museum.
- During the first ten years, productivity improvements of 20 to 30 percent were common. 2005 was a significant turning point.
- Two factories got a world-class accomplishment award from the Japan Institute of Plant Maintenance (JIPM) that year. In terms of TPM, this is the greatest level you can achieve.
- Unilever was an outstanding example of a company with this approach, with WCM based on TPM.

$$\text{WCM} = \text{TPM} + \text{LEAN} + \text{SIX SIGMA}$$



Mr. G. Prabu ram,
AP(SG)/MECH

[in](#) [Prabu ram Ganesamoorthy](#)



Mr. M. Thojesh Nandha,
(Batch 2018-2022)

[in](#) [Thojesh Nandha](#)

[ig](#) [thojesh_nandha](#)

REAL FOREST - DIGITISATION OF AGRO FOREST SAFETY

Raja Nursery, a horticulture based nursery garden, founded by K.S.KRISHNAMA RAJA & SONS in the year 1932. With vast experience in the nursery garden industry had supplied grafted saplings to agro farmers and agro land owners for about 89 successful years. Our major varieties of grafted mango plants are PANCHAVARNAM, RAJAPALAYAM SAPPATAI, MOHANDAS (PULIYADI), IMAMPASANTH, and ALPHONSO.

The year 2020 was a great year for us, as our mobile app Real Forest that digitises the agroforestry, has been acknowledged by Ramco Institute of Technology Research professors. My humble thanks for the MoU, that was signed with RIT in the year 2021, to have a technology research analysis and knowledge transfer for our product development.

Ramco family has been a mentor for us for decades, as our mango saplings has been planted in their family agro forest and our founders being the initial shareholders of Ramco cements.

As a 4th Generation nurserymen horticulturist and as a thanks giving to our agro farmer customers and agro land owners, who were been with us for 4 decades, we dedicated an agro forest based social networking mobile app named Real Forest, that digitises the agro forest by measuring trees and farm plantations, analyse the growth development of trees digitally with the help of Agriculture Universities, agro scientists, Farm expertise and mobile technologies to enrich the users to modern technology farming.

Real Forest app uses high end mobile technology to acquire tree data precisely. Together with the RIT research professors and RIT students, we are building the Real Forest app that enrich the digital data of agro forest trees with the following properties

- Protecting Agro forest through low cost mobile and digital surveillance system,
- Suggesting small agro farmers on plantation of area based Agro forest trees (Nativity Trees), based on AI learning from collected tree data through IOTs and Geographical data.
- Creating seamless Defi crypto receipt and payment to small agro farmers, backed up with their Tree Trusted digital data (Digitised Crypto Tree Volume) from their low cost mobile phones.
- Digitisation of Tree communication with waveforms and string theory concept, giving users digital data on Climate variations and Digital Disease control systems by digitising underneath root network structures.
- Enriching the life of small farmers and agro land owners (family forest) by accessing the benefits of Decentralised Tree Carbon Capture data that are block chained as Carbon Credit Crypto Token (niyantr) and NFT tokens.



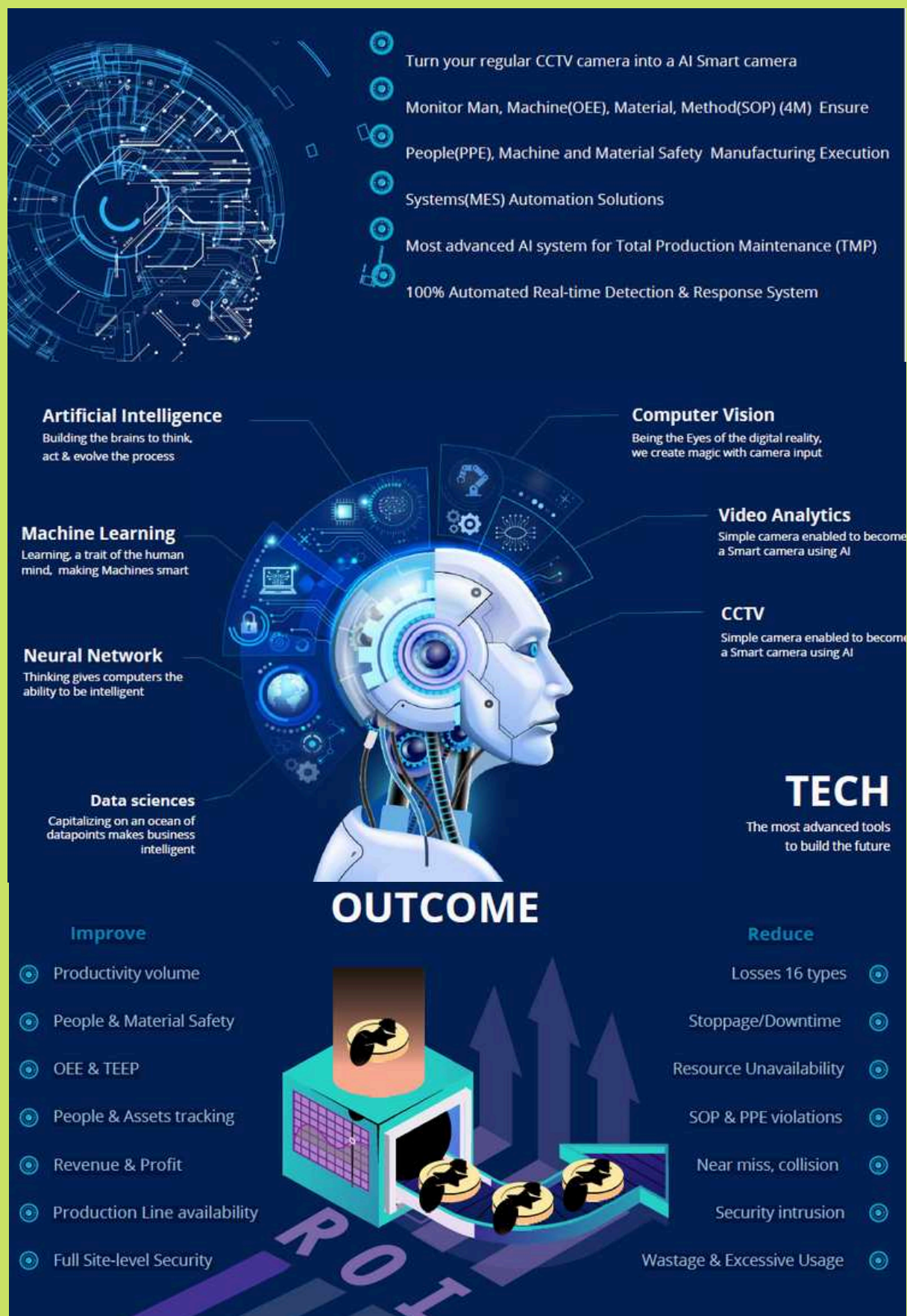
Mr. S. Aravind
Raja Nursery Garden,
134, T.P. Mills Road, Real Forest
Rajapalayam



REAL AI-POWERED SMART FACTORY SOLUTIONS

Mr. M. Deepak Kumar, Business Development Executive Robotronix Automation

COMPUTER VISION TRANSCENDING POSSIBILITIES OF INDUSTRY 4.0

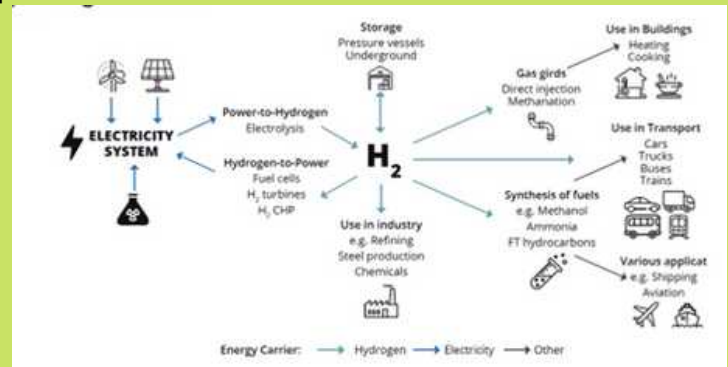
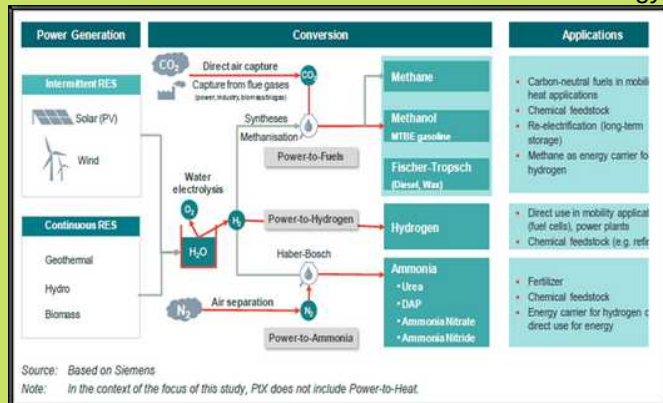


A NEW DECADE IN RENEWABLE ENERGY (HYDROGEN & POWER-TO-X)

Mr. P. Pavithran, AP/MECH

Apart from batteries, there are several new technologies that are being researched and developed for energy storage. These enabling technologies help stabilise energy systems' demand and supply issues by converting extra power into hydrogen, heat or some other form of energy carrier while at the same time decreasing the curtailment in renewable energy generation.

The big driver for these technologies is the availability of 'free' excess electricity from wind and solar power. Hydrogen is relatively easy and cheap to produce, but it has low energy density and thus is expensive to transport as such," explains Pitsinki. "Combustion technologies also have limited ability to use hydrogen. That is why the big push towards Power-to-X type of technologies, where hydrogen is further synthesised into methane or methanol that are easier to handle and use in energy production

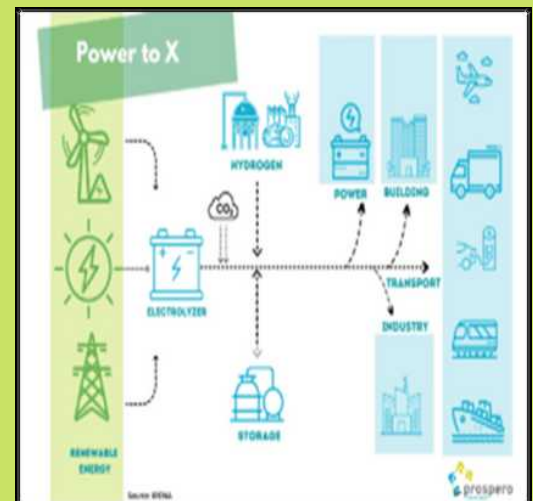


Wind and solar energy represent the main pillars of these plans, but hydrogen has a complementary role. It is seen as a way to enable large-scale integration of renewables into the power generation network; as a means of distributing energy across sectors and regions, and as a storage buffer to increase system resilience. Wind and solar require large-scale energy storage to compensate for short-term and seasonal imbalances. Due to the orders of magnitude involved, this can best be achieved by converting excess electricity using various Power-to-X concepts.

Hydrogen can be produced in numerous ways including fossil fuels, biomass, crops, nuclear energy and from renewable energy sources, such as wind, solar, geothermal and hydroelectric power. This diversity of potential supply sources is why hydrogen is seen as the ideal energy carrier. The main Power-to-X concept being targeted uses excess renewable electricity production from wind and sun to split water into hydrogen and oxygen in an electrolyser. As the hydrogen is produced from purely renewable sources, this is known as green hydrogen.

In the transition phase, natural gas reforming would be required. This is a carbon-neutral resource if carbon capture technology is utilized. The reforming of hydrocarbons is seen as a kick-starter to the provision of sufficient hydrogen supply at short notice, enabling the creation of the necessary hydrogen infrastructure. This includes investment into gas grids. Meanwhile, existing assets and infrastructure can be used with adapted gas turbine technology to reduce the carbon footprint of power generation and oil & gas. The scalability of gas turbines from small decentralized to large centralized systems offers enough flexibility in terms of production capability and local storage requirements.

Power-to-X concepts also offer an opportunity to reduce GHG emissions in heavy transportation vehicles, ships and air traffic. For instance, synthetic kerosene obtained using electricity from renewables is currently the only fuel enabling climate-neutral flight. Linking traditionally separated sectors of the energy system like electricity, gas, heat and transport, can raise energy efficiency and lower network investment costs. This is known as sector coupling. For example, coupling hydrogen gas turbines with other industries (e.g. chemicals and refineries) enables waste heat to be used



BREAKTHROUGH TECHNOLOGY FUELS INNOVATION IN FUTURE

Dr. P. Suresh Kumar, ASP/MECH

- Breakthrough Technology can be defined as those technologies that allow change at an accelerated speed. This transformation of the world is being facilitated by cost reductions and substantial progress in areas such as computing power, bandwidth, and data storage.
- The hierarchy and relationships between three exponential technologies such as Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning are represented using diagrammatic techniques.
- The list of Technology will nurture a multitude innovation and 21st century innovator must know these exponential technologies.
- Biotechnology is the use of scientific and engineering ideas to handle organic and inorganic materials in biological systems in order to produce goods and services. Products will be developed to slow down the ageing process, construct human parts, replace organs and tissues that have failed, and edit disorders in our DNA.
- Big data is a term that refers to large-scale data collections that are so complicated that they must be processed using non-traditional computer applications. Machine learning and deep learning are becoming increasingly popular as methodologies for handling enormous volumes of data, making big data and artificial intelligence projects indistinguishable.
- Blockchain is a mechanism that encrypts and links a growing list of information to safeguard your security and privacy. Blockchain is a distributed ledger that is maintained by numerous computers connected via a peer-to-peer network.
- Cybersecurity is concerned with the safeguarding of computing infrastructures and everything associated with them. The greatest threat to our privacy and security is the rise of big data and the power of machine learning algorithms, which are becoming more widely used.
- Quantum computing uses quantum mechanical events to manipulate data in the form of quantum bits or qubits. You can think about all of the various solutions to an issue and eliminate the ones that don't work.
- Virtual assistants are computer programmes that aid people by delivering information and automating chores. They will become the primary conduit for individuals to receive information, products, and services during the next decade.
- In the medical realm, Human Augmentation (cyborgs) is possibly the most well-known. Prostheses are mainly used to replace or complement bodies that have been injured or lost. However, by means of bionic implants it is possible to imitate the original function of the organs or parts of the body and even increase them.
- Automation is described as the technology that allows a process or procedure to be completed with little or no human intervention. Various research and jobs have proved that, as employment are lost, many new jobs will be established in hitherto undiscovered activities.
- Smart City is a concept that comes from an urban region that collects data using various types of electronic Internet of Things (IoT) sensors and then uses that data to efficiently manage assets and resources.
- Cloud computing refers to the provision of computing services on demand without the need for active user administration. Private, public, and hybrid clouds are the three sorts. Software as a service (SaaS), platform as a service (PaaS), and infrastructure as a service are examples (IaaS).
- Unmanned Aerial Vehicles (drones) come in a wide range of designs, sizes, and attributes. Drone data gathering capabilities, backed by cloud services and big data approaches, have the potential to change the competitive dynamics of the information landscape.
- Within the industrial realm, 3D printing has progressed from developing functional or aesthetic prototypes to manufacturing high-precision items. 3D printing is a collection of procedures in which a three-dimensional item is created by joining or solidifying materials under computer control.
- Computer telepathy suggests that capturing a thought and sharing it with the rest of the globe will be possible. Six scientists from Washington State University were able to connect three brains in order to play Tetris with the mind in 2019. BrainNet is the name of the system, and it functions as a direct cooperation interface between brains.
- A direct communication link between an enhanced or connected brain and an external device is known as a brain-computer interface. Through technological implants, we are enhancing our sensory, physical, and cognitive abilities.
- A computer-generated interactive experience that takes place in a simulated environment is classified as Virtual and Augmented Reality. Augmented reality is an interactive experience of a real-world environment in which computer-generated perceptual information is used to augment items in the actual world, sometimes through several sensory modalities.
- Autonomous vehicles are those that can mimic human handling and control abilities. They are aware of their surroundings and navigate accordingly. The level of automation at which the human driver becomes a passenger is level 5.

ELECTRIC VEHICLES – THE FUTURE OF MOBILITY

Mr. J. Jerold John Britto, AP(SG)/MECH

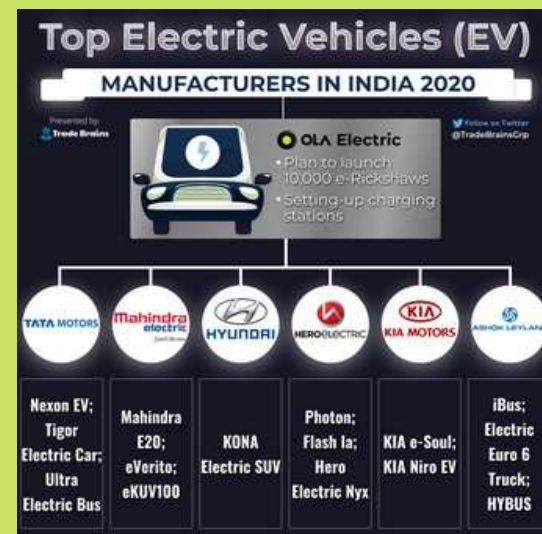
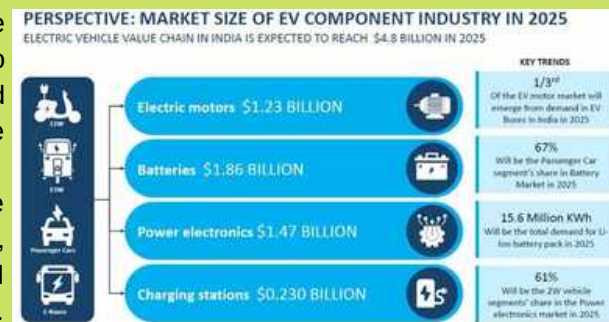
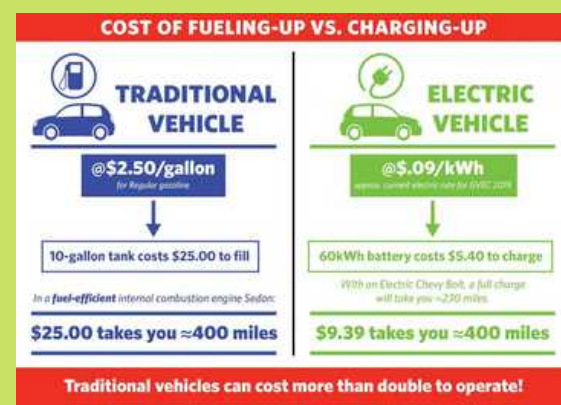
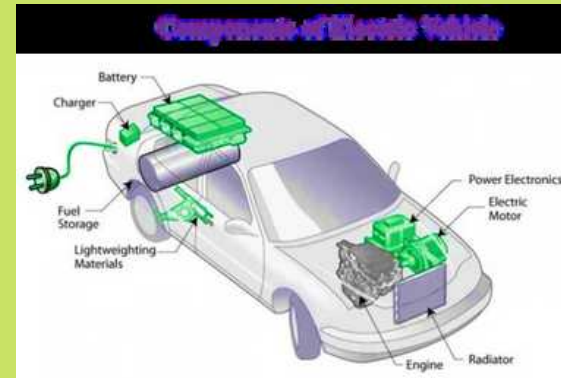
Electric Vehicles (EVs)

- An EV operates on an electric motor instead of an internal combustion engine and has a battery instead of a fuel tank.
- In general, EVs have low running costs as they have fewer moving parts and are also environmentally friendly.
- In India, the fuel cost for an EV is approximately 80 paise per kilometre. Contrast this with the cost of petrol which is today more than Rs 100 per litre in Indian cities, or Rs 7-8 per kilometre to operate a petrol-based vehicle.

What are the main Challenges to E-Mobility

- Electric Vehicles are expensive: Yes, its true, still the prices of electric vehicles are expensive for many of us. Automotive car OEMs are still in the infancy state of their product bathtub curve. In the coming years, when the technology matures and production increases, the price will come down.
- Battery technologies & performances: The attractiveness of an electric car stands or falls with the batteries: what distance does the car make with them, what do they cost, how much do they weigh? ...are the critical parameters for any EV; but unfortunately the battery technologies for EV still far behind to meet the mass-volume demands, in terms of volume and price.
- Infrastructure: Except a handful of countries, in rest of the world, the electrical grid and charging infrastructure is not ready to accommodate a massive intake of EVs. The existing electrical grid connections need a major renovation – which will be a huge investment, and in some cases it may not be possible at all.
- Customer Awareness: This is a hard truth; most of the people in the world still do not have enough awareness about electric mobility, nor any other green energy technologies for that matter. We still hear so many voices of fears around electric cars. Aren't we?. Common man/woman's willingness to participate is the key for the success of any new technology.

Electric Cars	
Pros	Cons
• Less noise pollution • No fumes • No direct greenhouse gas emissions • No air pollution • Perfect for short distances • Electric energy is cheaper compared to fossil fuels • Hedge against increasing gas prices in the future • Larger average storage opportunities • Used electric cars can be quite cheap • Less maintenance required • Government subsidies • Faster commuting • Could be charged at home	• Rather high costs for new cars • High depreciation in the first years • Battery may become weak over time • High costs for battery replacement • Low range • Charging may be an issue • Charging takes much longer • Outdated due to technological progress soon • Not eco-friendly if power comes from conventional energies • Limited selection of electric cars



AUTONOMOUS VEHICLE

Mr. M. Ramar, AP/MECH

Mr. R. Venkatesh , AP/MECH

“Most conversation to date has focused on the technological evolution of driverless cars – so it’s hugely encouraging to see the potential benefits that the technology enables are resonating with future passengers. Customer expectations of in-car experiences will not only impact the automotive industry, but other industries like media and entertainment, retail, and healthcare as well, paving the way for a plethora of collaborative business opportunities.”

In addition to their preferences on riding in a car, consumers believe autonomous cars will take on a larger role in daily life, going beyond just the act of driving. For example:

- 49% would be comfortable with self-driving cars running an errand on their behalf;
- 54% would trust an autonomous vehicle to drop off or pick up non-driving friends and family members;
- 50% expect self-driving cars to help them save time to pursue other activities, such as socializing, entertainment, working, or simply enjoying the journey.



Not so fast, friend

- While there is much optimism, the study also pointed out that barriers remain to adoption, including security and safety issues. The main barriers that could prevent consumers from adopting self-driving cars included:
- Self-driving vehicles getting confused by unexpected situations (71%)
- Vehicle security from hackers (73%)
- System security from hackers (72%)
- Difficulty interacting with human-driven vehicles (56%)
- Understanding new traffic rules (51%)
- Difficulty interacting with other self-driving vehicles (46%)
- Learning to use self-driven vehicles (37%)

AGILE MANUFACTURING

Mr. R. Venkatesh , AP/MECH

Mr. S. Kathiravan, AP/MECH

Agile Manufacturing was originally introduced with the publishing of a paper titled 21st Century Manufacturing Enterprise Strategy in the United States. As a result, in reaction to more competitive global marketplaces, it has been adopted as a technique of improving competitive advantage. The agile idea focuses on responding quickly to changing client demand. Flexibility is a critical component in agile production. To assure growth and success, agile manufacturing methods can be altered and merged into businesses of any size. In the contemporary world, there have been three significant phases or paradigm shifts in industrial production.

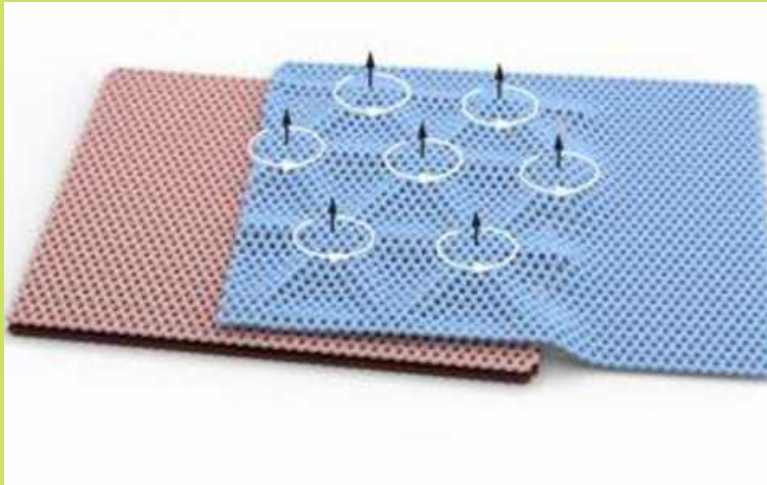
These phases are as follows:

- Craft production
- Mass production
- Lean/JIT production



RARE MAGNETISM FOUND IN THE WORLD'S STRONGEST MATERIAL

Dr. J. Jabinth, AP/MECH



A stacked and twisted set of graphene sheets leads to a rare form of magnetism, in which electrons across the stacks swirl in unison. (Image credit: Columbia University)

MAGNETISM

- Graphene, one of the world's strongest materials, isn't normally magnetic. But when stacked and twisted, graphene develops a rare form of magnetism, new research finds.
- The magnetic field isn't created by the usual spin of electrons within the individual graphene layers, but instead arises from the collective swirling of electrons in all of the three-layers of the stacked graphene structure, researchers reported Oct. 12 in the journal *Nature Physics*.
- Graphene is a material made of a single layer (or monolayer) of carbon atoms arranged in a honeycomb pattern. It's incredibly light and strong (though it is vulnerable to cracking). It also conducts electricity, making it exciting for use in electronics and sensors. It is found that by varying the number of graphene layers endows these composite materials with some exciting new properties that had not been seen before.
- When stacked two layers of graphene and then added a single layer on top, rotating the stack by 1 degree, then studied this graphene sandwich in a variety of circumstances, including temperatures just above absolute zero (the point at which all molecular motion stops). At these low temperatures, they found that the graphene stopped conducting electricity and became an insulator instead.
- They also found that they could control the properties of the twisty stack of graphene by applying an electric field. When the electric field was oriented in one direction, the system acted like a twisted double layer of graphene. When they reversed the field, the stack took on the properties of a twisted four-layer graphene structure.
- Perhaps strangest of all was the rare magnetism that appeared in the three-layer structure. A study published by another group in the journal *Advanced Materials* found that graphene bonded with boron nitride can give rise to a strange magnetic field; that field arose from the molecular bonds of the carbon in graphene and the boron in boron nitride. The new research reveals that this same type of magnetism can occur in pure graphene alone, simply because of interactions between carbon molecules.
- Pure carbon is not magnetic but we can engineer this property by arranging our three graphene sheets at just the right twist angles. The structure also contains regions where the properties are undisturbed by the twisting of the layer. These unique areas in the material could be exploited for data storage or quantum computing applications.
- The researchers are now planning to delve deeper into the fundamental properties of the graphene structure. "This is really just the beginning"

CRYOTHERAPY

Dr. M. Lakshmanan, AP(SG)/MECH

Cryosurgery Trusted Source is a type of surgery that involves the use of extreme cold to destroy abnormal tissues, such as tumours.

- The surgery most often involves the use of liquid nitrogen, although carbon dioxide and argon may also be used. When liquid nitrogen has a temperature between -346 and -320°F , it instantly freezes nearly anything that's in contact with it. In the case of human tissue, it can kill and destroy cells upon contact. This is important when the cells you want to kill are cancerous.
- Cryosurgery is typically used for tumours or precancerous lesions found on your skin. However, some tumours inside the body can be treated this way as well. Cryosurgery, also called cryotherapy, is similar to the technique used when doctors freeze off warts using liquid nitrogen spray.

Why cryosurgery is performed:

- Cryosurgery is used to destroy problem tissues in the body. In most cases of cancer, it's not the first line of defense.
- Cryosurgery is most often Trusted Source performed to treat cancers or precancerous lesions on the skin. It's also used on some internal organs, such as the liver, when disease and other problems make conventional surgery difficult or risky.
- Cryosurgery is used as the primary treatment Trusted Source for early prostate cancer that's contained in the prostate. It's also performed when cancer returns after other therapies have been used.

How a cryosurgery is performed?

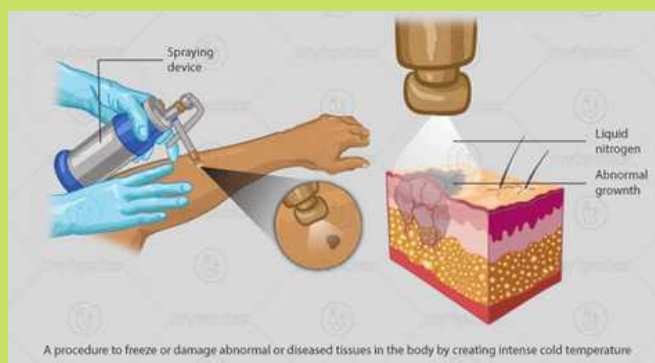
- Your doctor will Trusted Source place liquid nitrogen on your skin using a cotton swab or spray. A numbing medicine may be used to prevent pain or discomfort.
- If an internal area is being treated, your surgeon will use a scope, which is a flexible tube that can fit into various openings in your body, such as the urethra, rectum, or a surgical incision.
- The liquid nitrogen is fed to the area under treatment and applied to the targeted cells. The cells freeze, die, and then will be slowly absorbed by your body.
- Your doctor will use imaging equipment, such as an ultrasound, as a guide for carrying out the procedure.

Cryosurgery risks:

Cryosurgery does have risks, but they're considered lower than other cancer treatments, such as surgery and radiation.

The risks associated with cryosurgery include:

- Blisters
- Damage to nearby healthy tissue or vessels
- Infection
- A loss of sensation if nerves are affected
- Pain
- Scarring
- Ulcers
- Skin discoloration



RENEWABLE SECTOR: THE REDEEMER TO MEET ELECTRICITY DEMAND

Dr. V. Sivakumar, ASP/MECH

Mr. R. Arun Kumar, AP/MECH

Electricity is one of the primary driving forces in a country's economy. Its demand rises every day. India's average electricity demand ranges from 1,50,000 MW to 1,75,000 MW. The peak demand for electricity in India was recorded at 2,00,570 MW on July 7th, 2021. The total installed power capacity in India as on 31st March 2021 was 3,82,151 MW, with an annual growth potential of 3.25% against the previous year. In spite of a rapid increase in demand and faster depletion of fossil fuels, India still depends mostly on conventional fuels for electricity generation, with 61% of total demand relying on thermal-based power plants. As of April 1, 2021, approximately 32,285 MW of coal and gas-fired power plant projects are under construction. Among total installed capacity, 1,40,642 MW is contributed through renewable energy sectors, which include hydro, solar, and wind energy. It is estimated that the country's electricity demand will double by 2040 from its present demand

Global oil reserves deplete at a much faster rate than coal, which impacts the transportation sector. Based on the current reserve against production ratio, it is predicted that the oil reserves will exist only for the next 50 years. The depletion of crude oil forces countries to shift towards electric vehicles, which will increase the demand for electricity further. Every day, India imports 50,30,000 barrels* of crude and oil products, putting it in third place as an oil importer by 2020. When the oil reserves run out of crude, the energy requirements met by the above 5 million barrels will need an alternate source. As most of the imported oil is used for the transportation sector, the shortfall has to be met through electricity as electrical vehicles are going to be the alternate for conventional IC engine vehicles. At the Conference of Parties (COP) 29 Climate Summit held in Glasgow, India committed to achieve net zero emissions by 2070 and reducing the economy's carbon intensity to 45% by 2030. Further, India assured to 1 billion tonnes of carbon emissions from total projected emissions by 2030.

To maintain the commitment, harvesting renewable energy will be the only way to meet future energy demand. With a growth rate of over 20% in the renewable sector, India has committed to increase its renewable energy capacity to 500 GW by 2030, which is expected to meet 50% of the total energy requirement at the COP29 Climate Summit. With all these commitments and efforts, the scope of the renewable sector in India has wide openings in which students can work to build their knowledge and skills. This will make them contribute towards infrastructure establishment in the renewable sector, which will be the redeemer to meet the future electricity demand.

****One barrel is approximately 160 litres***

ROBOTIC PROCESS AUTOMATION

Mr. S. Valai Ganesh, AP/MECH

Robotic Process Automation (RPA) is a rapidly growing technology that is revolutionizing the way businesses operate. It involves the use of software robots to automate repetitive, manual tasks, allowing employees to focus on more strategic and creative work.

RPA has many benefits, including increased efficiency, cost savings, and improved accuracy. It is especially useful for industries that rely heavily on repetitive tasks, such as finance, healthcare, and manufacturing.

One of the key advantages of RPA is that it can work 24/7 without needing breaks or downtime. This means that tasks can be completed faster and more efficiently, leading to increased productivity and faster turnaround times.

Another advantage of RPA is that it can significantly reduce errors and inconsistencies in data processing. The software robots are programmed to follow specific rules and procedures, which helps ensure that data is processed accurately and consistently every time.

RPA can also free up valuable time for employees, allowing them to focus on higher-level tasks that require more creativity and problem-solving skills. This can help increase job satisfaction and engagement, leading to a more motivated and productive workforce.

Implementing RPA requires careful planning and coordination with business stakeholders. It is important to identify which processes are best suited for automation and to ensure that the software robots are properly integrated into the existing IT infrastructure.

There are many different RPA tools and platforms available on the market, each with its own strengths and weaknesses. It is important to choose the right tool for the specific needs of the business and to ensure that it is compatible with existing systems and workflows.

Overall, RPA is a powerful tool for businesses looking to improve efficiency, reduce costs, and increase productivity. By automating repetitive and manual tasks, RPA allows employees to focus on more strategic work, leading to improved job satisfaction and engagement. With careful planning and implementation, RPA can be a game-changer for businesses across a wide range of industries.



DIFFERENCE BETWEEN TRADITIONAL ROBOTS VS. COBOTS

Mr. T. Selva Sundar, AP/MECH

TRADITIONAL	COBOTS	COBOT EOAT
Big batches, little variability Ideal for large companies that manufacture high volumes of the same products for long periods Indicator goes here	Low-volume, high-mix Designed for low-volume, high-mix production, where the robot is often redeployed for new processes	Easy to change Flexible, quick-change tooling to eliminate downtime between processes
Complex deployment Requires extensive programming skills and takes days or weeks to set up	Fast and easy deployment Easy to deploy with simple programming that inexperienced users can set up in minutes	Easy to program Tooling that is fast and easy to program and deploy using the cobot's built-in teach environment
Requires constancy Programmed for unchanging environment and the same movement with minimal need to adapt	Adapts to environment Flexible to adapt to changing environment and work pieces to be handled	Adaptable Tools that easily adapt to varying sizes, shapes, and conditions of workplaces and the environment
Not safe without guarding Typically requires safety guarding to keep human workers out of the robot's work cell	Collaborative and safe After risk assessment, humans can work alongside robot in collaborative applications	Design for safety Safe, collaboratively-designed tools that simplify interaction with humans
Focus on the robot Repeats the same actions for years, with unchanging tool that is integrated for a specific process	Focus on the EOAT As robot arm becomes a commodity, focus shifts to EOAT to increase robot utilization	Flexible Flexible tooling that can be used for multiple processes

மானுட வாழ்வு

ஐந்தறிவு போதுமென்று
 அந்தியிலே சாய்ந்திடும்
 ஆங்கோர் விலங்கினை போல்
 வாழ்வதன்று மானுட வாழ்வு!
 ஆறாம் அறிவினை அறிந்தோம்
 என்பதை அறிவுறுத்த
 அறிந்து பெருக்கிடு உன் அறிவினை
 அதன்பின் உணர்த்திடு உலகினுக்கே!!

ம.அசோக் குமார்
 உதவிப் பேராசிரியர்
 (முதுநிலை)
 இயந்திரவியல் துறை



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